

02/09/25

Expt. No. 2

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Compare the strength of Acids in a Mixture of Acids using Conductivity Meter

Aim:-

To estimate the amount of hydrochloric (HCl) and acetic acids (CH_3COOH) present in the given mixture. You are provided with a standard solution of NaOH of strength 0.2N.

Materials Required:-

Conductivity Meter, Conductivity cell, burette, pipette, std. 0.2N NaOH.

Principle:-

Conductivity of a solution depends up on the number and nature of ions in solution. During addition of base to acid solution, the H^+ ions are removed as water. Hence the conductance the solution decreases.



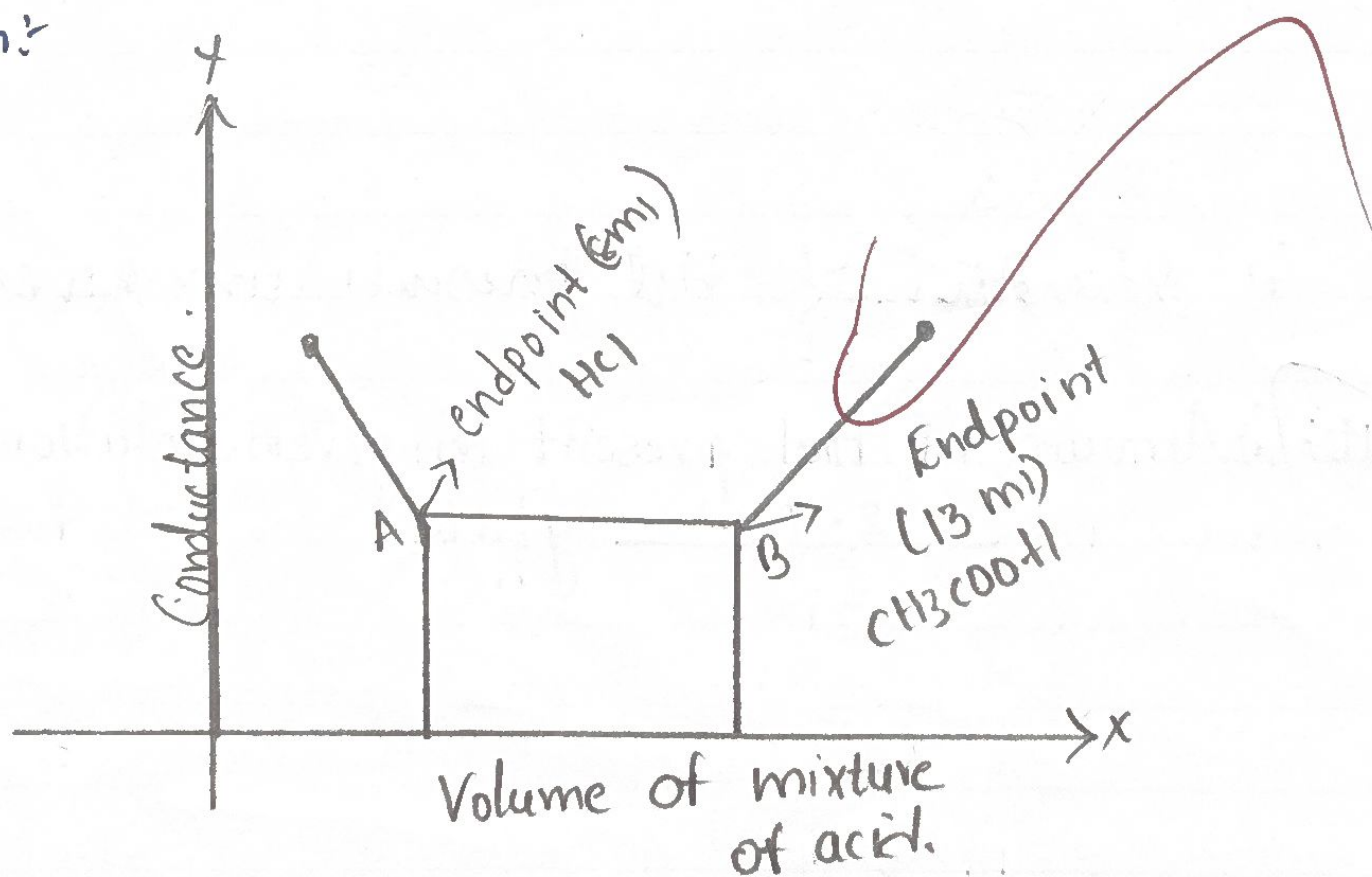
acid base.

In a mixture of strong base and weak acid. strong acid reacts with the base initially. The decrease in conductivity is very sharp. After the neutralisation of strong acid, weak acid reacts with the added base. At the end point, when all the acids are neutralized, the addition of base increases the OH^- ion in the solution. Hence there is sharp increase in conductivity.

Short procedure

SNO	Content	Titration: Mixture of Acids vs strong Base.
1.	Burette Solution.	Standard NaOH
2.	Pipetter Solution.	20ml of mixture of acids (HCl + CH ₃ COOH)
3.	Additional solution.	20ml of Conductivity water
4.	Indicator.	—
5.	Endpoint	Decrease in Conductance and Increase in Conductance.
6.	Formula	$V_1N_1 = V_2N_2$
7.	Equivalent of	HCl = 36.45
		CH ₃ COOH = 60.

Model graph:-

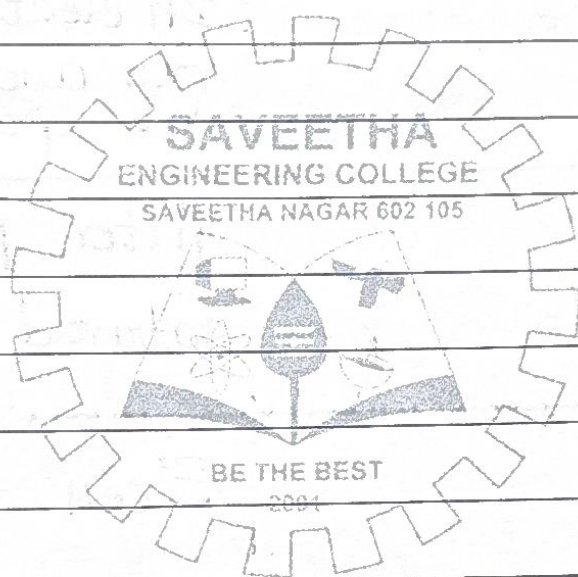


Procedure:-

The burette is filled with Sodium Hydroxide (NaOH) Solution up to zero mark. 20 ml of mixture is taken in a 100 ml beaker. The conductivity cell is placed in it and it then diluted to 20 ml with distilled water. It is then connected with the conductivity meter. Then 1 ml of alkali is added from the burette to the solution. The solution is stirred carefully and then conductance is measured after each added alkali.

A graph is drawn conductance and volume of NaOH . From the graph the first end point A and second point end point B are noted. The amount of HCl and CH_3COOH present in 1 litre of the mixture is calculated from the endpoints A and B respectively.

SNO	Volume of NaOH added (ml)	Conductance (cm) (L ⁻¹)
1	0	4.03.
2	1	4.12
3	2	4.20
4	3	4.78
5	4	8.95
6	5	8.87
7	6	2.71
8	7	2.35
9	8	2.22
10	9	3.36
11	10	2.40
12	11	2.58
13	12	3.20
14	13	3.20
15	14	3.21
16	15	3.71
17	16	4.00
18	17	4.20
19	18	4.36
20	19	4.86.
21	20	



~~Result:~~ 1. The amount of HCl present in litre of the given solution = 2.912 gl.

2. The amount of CH_3COOH present in litre of the given solution = 3.0 gl.

Calculation of strength of HCl:

Volume of mixture (HCl) $V_1 = 20 \text{ ml}$

Strength of mixture (HCl) $N_1 = 0.08 ?$

Volume of NaOH $V_2 = 8 (A) \text{ ml}$

Strength of NaOH $N_2 = 0.2 N$

$$N_1 = V_2 N_2 / V_1 = V_2 \times 0.2 \times 20$$

strength of HCl

The amount of HCl present

$$\begin{aligned} \text{in 1 litre of the given solution} &= \text{strength of HCl} \times \text{Eq. wt of HCl} \\ &= \frac{0.08}{1} N \times 36.5 \\ &= \underline{2.92} \text{ g/l} \end{aligned}$$

Calculation of strength of CH_3COOH

Volume of mixture (CH_3COOH) $V_1 = 20 \text{ ml}$

Strength of mixture (CH_3COOH) $N_1 = \underline{\hspace{2cm}} ?$

Volume of NaOH (B-A) $V_2 = \underline{5} \text{ ml}$

Strength of NaOH $N_2 = 0.2 N$

Strength of CH_3COOH $N_1 = \underline{0.05} N$

The amount of CH_3COOH present in.

$$\begin{aligned} \text{1 litre of given solution} &= \text{strength of } \text{CH}_3\text{COOH} \times \\ &\quad \text{Eq. wt of } \text{CH}_3\text{COOH} \end{aligned}$$

$$\begin{aligned} &= \frac{0.05}{1} N \times 60 \\ &= \underline{3.0} \text{ g/l.} \end{aligned}$$